Analysis Report

AgroFlow:

Agricultural Equipment Sharing Platform

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1. Introduction

AgroFlow is a digital platform that allows farmers in Pakistan to rent agricultural equipment from equipment owners. It solves the problem of small farmers who cannot afford to buy expensive machinery, and equipment owners whose machines sit unused most of the year.

Key Aspect	Details
Platform Type	Web and Mobile Application with Offline Support
Primary Users	Farmers, Equipment Owners, Administrators
Phase 1 Features	Booking, KYC Verification, Escrow Ledger, Photo Inspections
Phase 2–3 Features	IoT Tracking, AI Demand Forecasting, Computer Vision
External Systems	NADRA (identity), IoT Device (tracking), FBR (tax reporting)

2. User Scenarios

Each use case below describes one user interaction with the AgroFlow system. It includes who is involved, what they want to achieve, the steps taken, and what happens if something goes wrong.

UC-01: Register & Verify Account

Actors	Farmer, Equipment Owner | NADRA System (External)
Goal	User creates a verified account on AgroFlow using their CNIC.
Preconditions	User has a valid CNIC. Equipment Owners also need a Fard document.
Trigger	User opens the app and taps Register.
Main Steps	1. User selects role: Farmer or Equipment Owner. 2. User enters name, phone number, and district. 3. User uploads CNIC photos. Equipment Owners also upload Fard document. 4. System sends CNIC to NADRA for identity verification. 5. NADRA confirms, identity account is created and activated. 6. User receives confirmation and can log in.
Alternative Flows	• CNIC photo is blurry → system asks user to re-upload. • NADRA verification fails → account set to Pending; Admin reviews manually. • NADRA unavailable → request is queued; user is informed.
Postconditions	Account created with Verified status. Role assigned. Event logged.
Priority	High

UC-02: List Equipment for Rent

Actors	Equipment Owner
Goal	Owner adds equipment to the platform so farmers can find and book it.
Preconditions	Owner is registered and KYC verified.

Trigger	Owner taps Add Equipment from the dashboard.
Main Steps	1. Owner enters equipment type, make, model, and year. 2. Owner enters horsepower, fuel type, and last service date. 3. Owner sets daily rental price and available dates. 4. Owner uploads at least 3 photos of the equipment. 5. Owner sets home location on the map. 6. System creates the listing and generates a QR code. 7. Equipment appears in farmer search results.
Alternative Flows	• Photo quality too low → system asks for better photos. • GPS not available → owner selects location manually.
Postconditions	Equipment listed as Available. QR code generated. Event logged.
Priority	High

UC-03: Search & Book Equipment

Actors	Farmer
Goal	Farmer finds available equipment nearby and sends a booking request.
Preconditions	Farmer is KYC verified. At least one listing exists in the area.
Trigger	Farmer taps Find Equipment on the home screen.
Main Steps	1. Farmer selects equipment type and district/tehsil. 2. System shows nearby available equipment sorted by distance. 3. Farmer views results showing distance, daily rate, and photos. 4. Farmer selects equipment and reviews full details. 5. Farmer picks rental dates and confirms the booking. 6. Booking is created in REQUESTED state. 7. Owner receives a notification to confirm or reject.
Alternative Flows	• No equipment found → system expands search area. • Dates conflict → system shows next available window.
Postconditions	Booking created. Escrow entry created. Owner notified.
Priority	Critical
Includes	UC-05 (Pre-Rental Inspection), UC-08 (Update Escrow Ledger)

UC-04: Confirm / Reject Booking

Actors	Equipment Owner
Goal	Owner reviews a farmer's request and confirms or rejects it.
Preconditions	Booking exists in REQUESTED state.
Trigger	Owner opens the booking notification.
Main Steps	1. Owner reviews booking: dates, farmer profile, location. 2. Owner taps Confirm, booking moves to CONFIRMED state. 3. Pre-rental inspection is scheduled. 4. Farmer receives confirmation notification.
Alternative Flows	• Owner taps Reject → booking cancelled; farmer notified. • Owner does not respond in 24 hours → booking auto-cancelled.
Postconditions	Booking confirmed. Both parties notified. Escrow entry active.
Priority	High

UC-05: Pre-Rental Inspection

Actors	Equipment Owner, Farmer
Goal	Record equipment condition before handover to protect both parties.
Preconditions	Booking is in CONFIRMED state.
Trigger	Owner taps Start Inspection when equipment is about to be handed over.
Main Steps	1. Owner photographs equipment from at least 4 angles. 2. Owner records fuel level and current engine hours. 3. Owner records location of handover. 4. Farmer reviews photos in the app and signs off. 5. System saves the inspection report linked to the booking. 6. Booking moves to IN PROGRESS state.
Alternative Flows	• Farmer disputes condition → dispute raised before handover. • No internet → app works offline and syncs when restored.
Postconditions	Pre-rental inspection report saved. Booking set to IN PROGRESS.
Priority	High

UC-06: Track Equipment via IoT

Actors	IoT Device (Phase 2)
Goal	Monitor equipment location and usage during the rental period.
Preconditions	Equipment has an IoT device attached. Booking is IN PROGRESS.
Trigger	Equipment engine is started during active rental.
Main Steps	1. IoT device sends GPS location, engine hours, and fuel level. 2. System records this data against the active booking. 3. Owner can track equipment location in real time. 4. System alerts owner if equipment leaves the agreed work area.
Alternative Flows	• Equipment leaves geofence → Owner and Admin are alerted immediately.
Postconditions	Tracking data recorded. Usage stored for maintenance reporting.
Priority	Medium (Phase 2 — not in Version 1.0)
Extends	UC-03 (Search & Book Equipment)

UC-07: Post-Rental Inspection

Actors	Equipment Owner, Farmer
Goal	Record equipment condition after rental to check for any new damage.
Preconditions	Rental period is over. Booking is in PENDING RETURN state.
Trigger	Owner taps Start Return Inspection when equipment is returned.
Main Steps	1. Owner photographs equipment from the same angles as before rental. 2. Owner records current fuel level and engine hours. 3. Farmer reviews the post-rental photos and signs off. 4. Both parties confirm the return condition. 5. Booking moves to COMPLETED state. 6. Escrow payment is released to the Owner.
Alternative Flows	• Owner claims new damage → dispute is raised before closing booking.
Postconditions	Post-rental inspection saved. Booking completed. Escrow released.
Priority	High

UC-08: Update Escrow Ledger

Actors	Equipment Owner, Farmer
Goal	Track payment status securely without handling real money.
Preconditions	Booking has been confirmed.
Trigger	Included automatically when booking is confirmed.
Main Steps	1. System creates an escrow ledger entry in HELD state. 2. Farmer confirms external payment was made. 3. Owner confirms payment was received. 4. When rental completes with no dispute → escrow entry set to RELEASED. 5. Transaction record is saved permanently.
Alternative Flows	• Dispute raised → escrow stays HELD until Admin resolves it.
Postconditions	Escrow ledger updated. Financial record saved.
Priority	High
Note	No real money moves through the system — only payment status is tracked.

UC-09: Raise Dispute

Actors	Farmer / Equipment Owner
Goal	Formally report a problem with a booking.
Preconditions	An active or recently completed booking exists.
Trigger	User taps Raise Dispute on a booking.
Main Steps	1. User selects dispute category: damage, payment, performance, or non-return. 2. User writes a description and uploads supporting photos. 3. System notifies the other party. 4. Other party submits their response within 48 hours. 5. If unresolved → dispute is sent to Admin for a decision.
Alternative Flows	• Other party does not respond in 48 hours → dispute auto-sent to Admin.
Postconditions	Dispute record created. Escrow held. Both parties notified.
Priority	High
Extends	UC-10 (Resolve Dispute)

UC-10: Resolve Dispute

Actors	Administrator
Goal	Admin reviews evidence and makes a fair decision on the dispute.
Preconditions	Dispute has been escalated from UC-09.
Trigger	Admin receives dispute escalation notification.
Main Steps	1. Admin opens the dispute and reviews all evidence. 2. Admin reads inspection photos, booking history, and both parties' statements. 3. Admin makes a ruling: in favour of Farmer, Owner, or partial. 4. System updates escrow ledger based on the ruling. 5. Both parties are notified of the outcome.
Alternative Flows	• Admin needs more evidence → both parties given 24 hours to submit more.

Postconditions	Dispute closed. Escrow released per ruling. Audit record created.
Priority	High

UC-11: Cancel Booking

Actors	Farmer, Equipment Owner
Goal	Cancel a booking that is no longer needed.
Preconditions	Booking exists in REQUESTED or CONFIRMED state.
Trigger	User taps Cancel Booking.
Main Steps	1. User selects the booking and taps Cancel. 2. User provides a reason for cancellation. 3. System checks cancellation policy based on timing. 4. Booking is cancelled. Other party is notified. 5. Escrow entry is voided if payment was not yet made.
Alternative Flows	• Cancellation within 2 hours of rental start → cancellation fee may apply.
Postconditions	Booking cancelled. Both parties notified. Escrow voided.
Priority	Medium

UC-12: Approve KYC Verification

Actors	Administrator | NADRA System (External)
Goal	Admin reviews and approves a user's identity documents.
Preconditions	User has submitted registration and NADRA result is received.
Trigger	Included automatically after UC-01 registration.
Main Steps	1. Admin opens the KYC review queue. 2. Admin views CNIC details and NADRA response. 3. For Equipment Owners — Admin reviews the Fard document. 4. Admin taps Approve → account fully activated. 5. User is notified and can access all features.
Alternative Flows	• Fard document unclear → Admin manually reviews uploaded image. • Admin rejects KYC → user notified with reason.
Postconditions	KYC status set to Verified. User can use all platform features.
Priority	High

UC-13: View Analytics & Reports

Actors	Administrator, Equipment Owner
Goal	View platform statistics and generate reports.
Preconditions	Actor is logged in. At least one completed booking exists.
Trigger	Actor opens the Analytics section.
Main Steps	1. Actor selects report type and date range. 2. System shows charts: bookings, revenue, utilization rate, dispute rate. 3. Actor can export report as PDF or Excel. 4. Admin can send FBR compliance report to the government portal.
Alternative Flows	• No data for selected filters → system suggests widening the date range.

Postconditions	Report generated. Export event logged.
Priority	Medium

UC-14: Manage Users

Actors	Administrator
Goal	Admin manages all user accounts on the platform.
Preconditions	Admin is logged in with administrator access.
Trigger	Admin opens the User Management panel.
Main Steps	1. Admin searches users by name, CNIC, or region. 2. Admin views user profile, KYC status, and rental history. 3. Admin can approve or reject pending KYC submissions. 4. Admin can suspend or ban users who violate platform rules. 5. All admin actions are automatically saved in the audit log.
Alternative Flows	• Admin suspends user with active booking → booking cancelled; other party notified.
Postconditions	User status updated. Affected user notified. Action recorded in audit log.
Priority	High

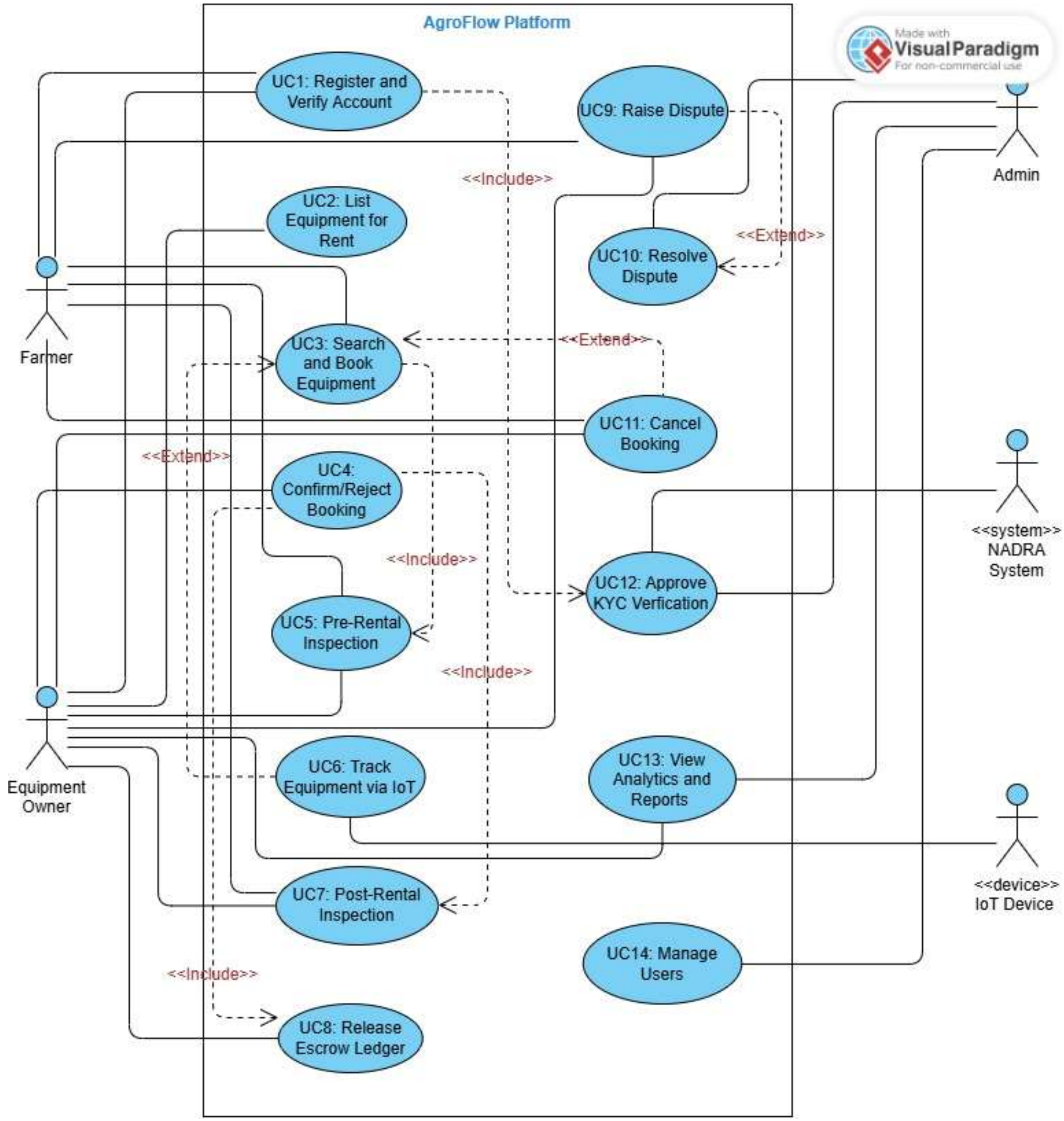
3. Use Case Diagram (UCD)

The Use Case Diagram below shows all actors and their relationships to the 14 use cases of AgroFlow. It was created in Visual Paradigm.

3.1 Actors

Actor	Type	Role
Farmer	Primary	Searches, books, and uses equipment. Pays via escrow.
Equipment Owner	Primary	Lists equipment, confirms bookings, conducts inspections.
Administrator	Secondary	Approves KYC, resolves disputes, manages platform.
NADRA System	External System	Verifies user identity via CNIC.
IoT Device	External (Phase 2)	Sends real-time GPS and engine data during rental.

3.2 Use Case Diagram



AgroFlow Platform
Made with Visual Paradigm For non-commercial use
UC1: Register and Verify Account
UC2: List Equipment for Rent
UC3: Search and Book Equipment
UC4: Confirm/Reject Booking
UC5: Pre-Rental Inspection
UC6: Track Equipment via IoT
UC7: Post-Rental Inspection
UC8: Release Escrow Ledger
UC9: Raise Dispute
UC10: Resolve Dispute
UC11: Cancel Booking
UC12: Approve KYC Verfication
UC13: View Analytics and Reports
UC14: Manage Users
<<Include>>
<<Extend>>
<<Extend>>
<<Extend>>
<<Include>>
<<Include>>
<<Include>>
Farmer
Equipment Owner
Admin
<<system>> NADRA System
<<device>> IoT Device

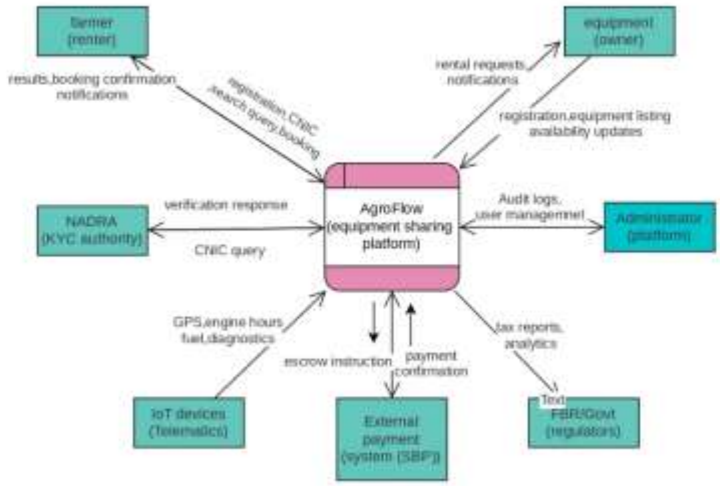
4. Data Flow Diagrams (DFD)

4.1 Context Diagram (Level 0)

The Context Diagram shows the entire AgroFlow system as one single process. It shows all external parties and what data goes in and out of the system.

External Party	Data INTO System	Data OUT of System
Farmer	Registration, CNIC, booking requests, dispute submissions	Search results, booking confirmations, payment receipts
Equipment Owner	Asset details, booking decisions, inspection data	Booking requests, escrow notifications, analytics
Administrator	KYC decisions, dispute rulings	Pending KYC queue, dispute cases, audit reports
NADRA	CNIC verification result	CNIC verification request
IoT Device (Ph.2)	GPS, engine hours, fuel data	Device configuration updates
FBR / Government	None	Tax reports, mechanization data

4.2 Context Diagram

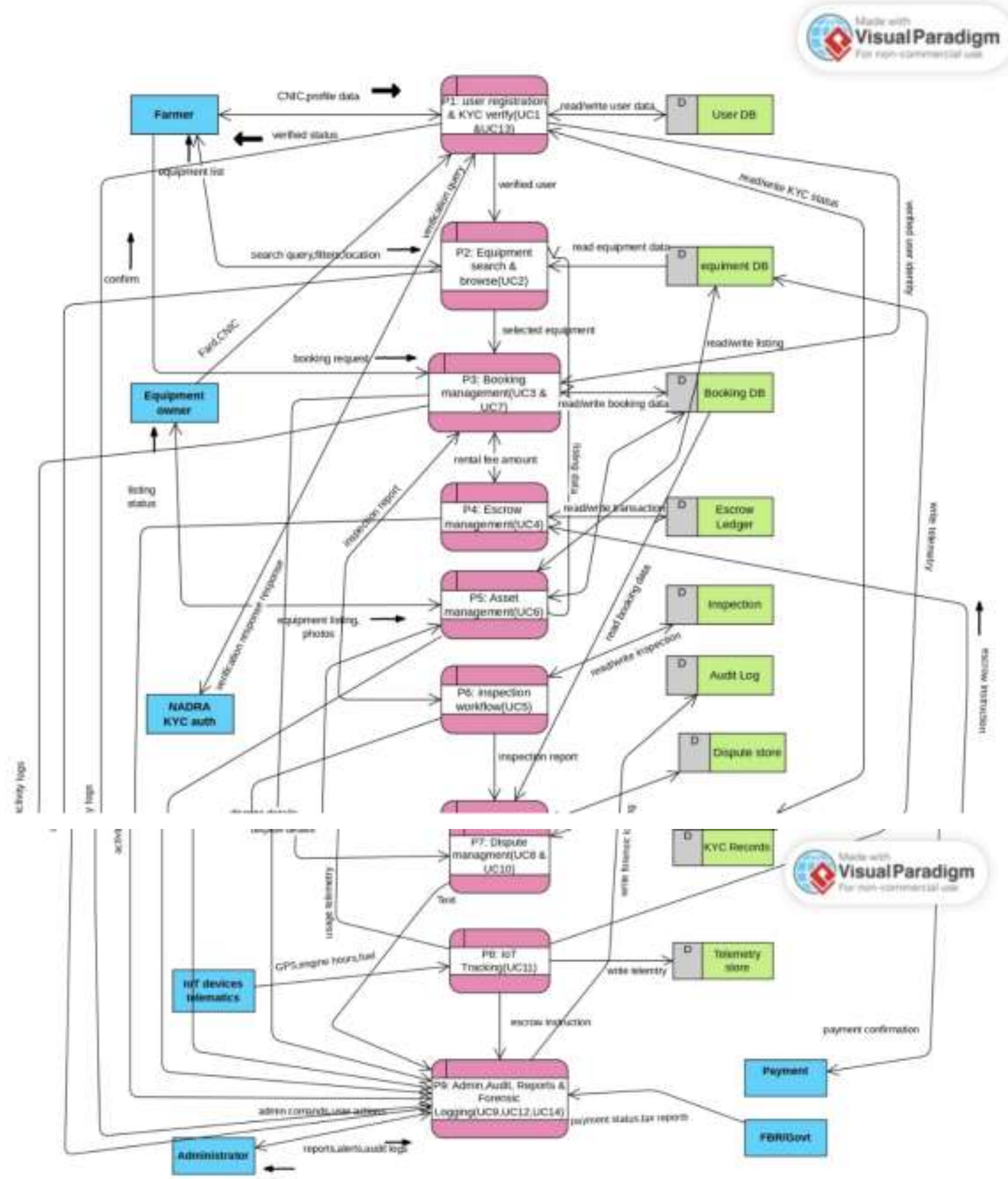

farmer
(renter)
results,booking confirmation
notifications
registration,CNIC
search query,booking
equipment
(owner)
rental requests,
notifications
registration,equipment listing
availability updates
NADRA
(KYC authority)
verification response
CNIC query
AgroFlow
(equipment sharing
platform)
Audit logs,
user managemnet
Administrator
(platform)
GPS,engine hours
fuel,diagnostics
escrow instruction
payment
confirmation
tax reports,
analytics
Text
IoT devices
(Telematics)
External
payment
(system (SBP))
FBR/Govt
(regulators)

4.3 Level 1 DFD — Main Processes

The Level 1 DFD breaks the system into 9 main processes and 9 data stores.

P#	Process	Key Inputs	Key Outputs
P1	**User Registration & KYC**	CNIC, Fard, role	Verified user profile
P2	**Equipment Search & Browse**	Search query, location	Equipment list shown to Farmer
P3	**Booking Management**	Booking request, dates	Booking record saved
P4	**Escrow Management**	Rental fee, booking completion	Payment release or refund
P5	**Asset Management**	Equipment listing, photos	Asset record saved
P6	**Inspection Workflow**	Photos, fuel level, engine hours	Inspection report saved
P7	**Dispute Management**	Evidence, admin ruling	Dispute record saved
P8	**IoT Tracking (Phase 2)**	GPS, engine hours via MQTT	Telemetry data saved
P9	**Admin, Audit & Reports, Forensics Loggin**	Admin actions, all system events	Reports, audit logs

4.4 Level 1 DFD



Made with Visual Paradigm For non-commercial use
Farmer
CNIC,profile data
verified status
P1: user registration & KYC verify(UC1 &UC13)
read/write user data
User DB
verified user
read/write KYC status
verified user identity
equipment list
search query,filters,location
P2: Equipment search & browse(UC2)
read equipment data
equiment DB
selected equipment
read/write listing
Farid,CNIC
booking request
P3: Booking management(UC3 & UC7)
read/write booking data
Booking DB
Equipment owner
confirm
listing status
rental fee amount
listing data
P4: Escrow management(UC4)
read/write transaction
Escrow Ledger
write telemetry
inspection report
verification query
verification response response
P5: Asset management(UC6)
equipment listing, photos
read booking data
Inspection
read/write inspection
NADRA KYC auth
P6: Inspection workflow(UC5)
Audit Log
inspection report
Dispute store
escrow instruction
P7: Dispute managment(UC8 & UC10)
KYC Records
Text
usage telemetry
P8: IoT Tracking(UC11)
GPS,engine hours,fuel
IoT devices telematics
write telemetry
Telemetry store
write finance
escrow instruction
payment confirmation
Payment
P9: Admin,Audit, Reports & Forensic Logging(UC9,UC12,UC14)
admin comands,user actions
payment status,tax reports
FBR/Govt
Administrator
reports,alerts,audit logs

4.5 Level 2 DFDs — Detailed Process Decomposition

Level 2 DFDs break each main process from Level 1 into smaller sub-processes. They show exactly how data flows inside each process. Four Level 2 diagrams are included below.

4.5.1 P1 — User Management & KYC Verification

This diagram shows how a user registers on AgroFlow and how their identity is verified through NADRA. It has 5 sub-processes:

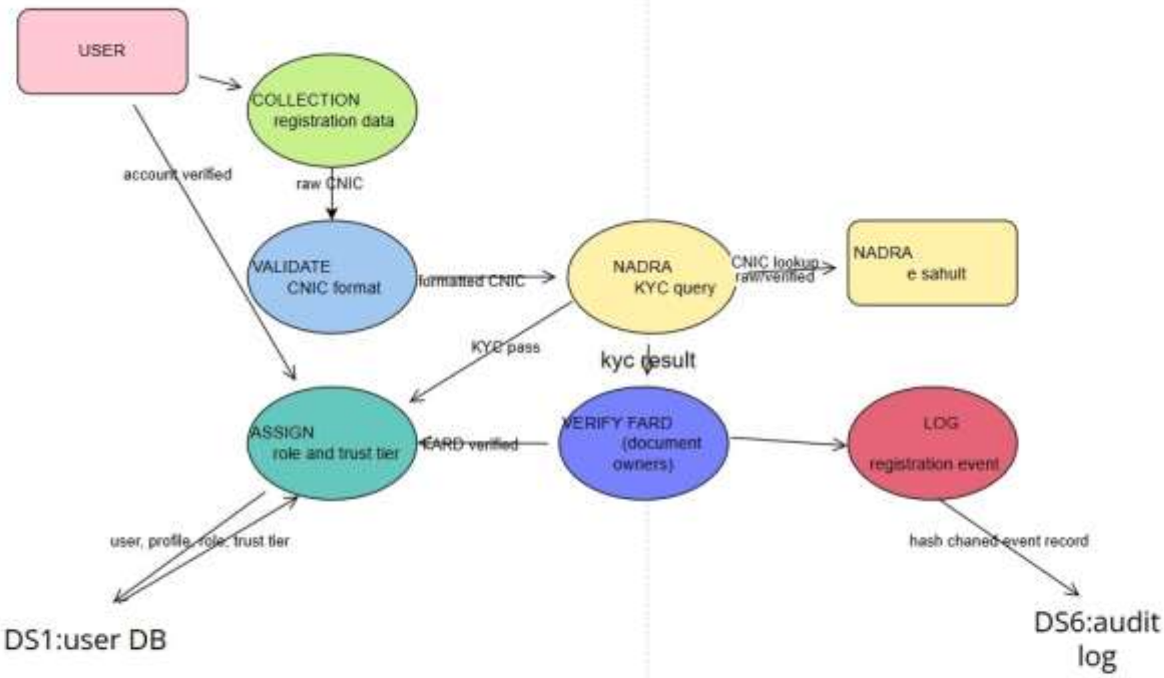
Sub-Process	What it Does	Input	Output
1.1 Collect Registration Data	Receives name, phone, district, and role from the user	User inputs	Raw registration data
1.2 Validate CNIC Format	Checks CNIC number format before sending to NADRA	Raw CNIC	Formatted CNIC
1.3 NADRA KYC Query	Sends CNIC to NADRA e-Sahulat API for identity check	Formatted CNIC	KYC result (pass/fail)
1.4 Verify Fard Document	Checks Fard land record for Equipment Owners only	Fard document	Fard verified/rejected
1.5 Assign Role & Trust Tier	Creates account, assigns role, sets initial trust score	KYC result, Fard result	User profile saved to DS1
1.6 Log Registration Event	Records the registration event in the audit log	User profile	Hash chain event → DS6

Data Stores used: DS1 (User DB), DS6 (Audit Log)

External Entities: User (Farmer/Owner), NADRA e-Sahulat System

P1 Level 2 DFD

LEVL 2: user managment and KYC verification



4.5.2 P3 — Booking Management

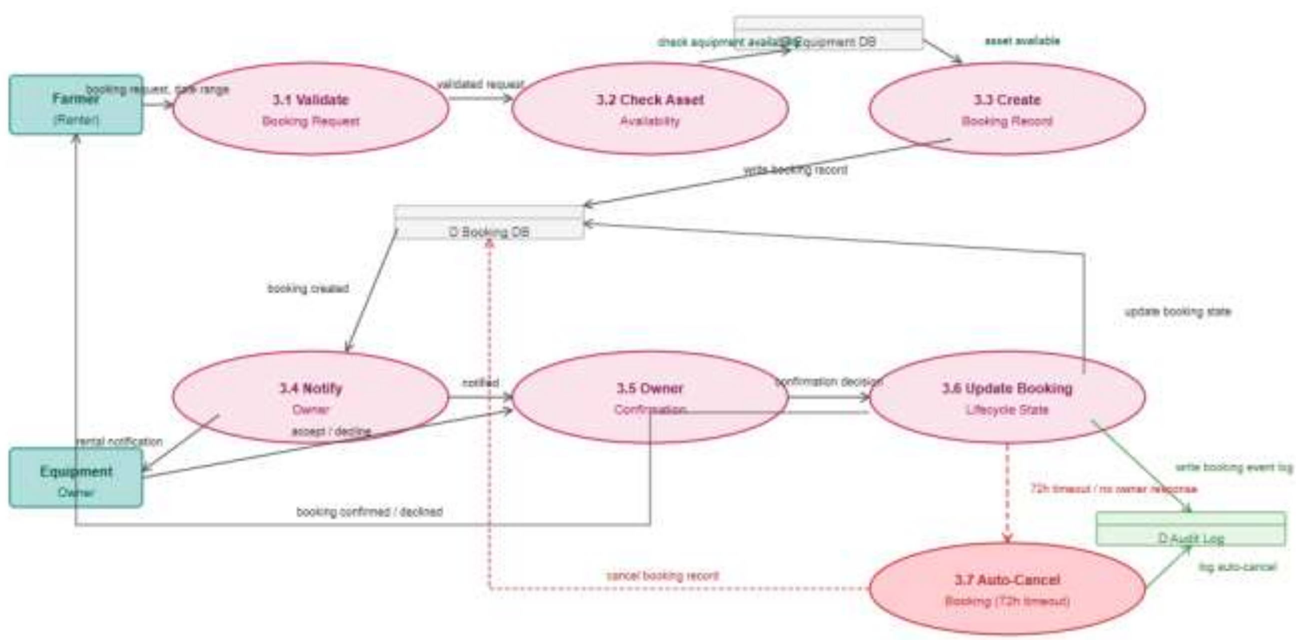
This diagram shows how a booking request is processed from submission to confirmation or auto-cancellation. It has 7 sub-processes:

Sub-Process	What it Does	Input	Output
3.1 Validate Booking Request	Checks farmer's KYC status and booking details	Booking request, date range	Validated request
3.2 Check Asset Availability	Checks if equipment is available for the requested dates	Validated request	Availability result
3.3 Create Booking Record	Creates a booking record in REQUESTED state	Available asset + dates	Booking record → Booking DB
3.4 Notify Owner	Sends booking notification to Equipment Owner	Booking created	Rental notification to Owner
3.5 Owner Confirmation	Owner accepts or declines the booking	Accept / decline decision	Confirmation status
3.6 Update Booking Lifecycle	Updates booking state based on owner's decision	Confirmation status	Booking state updated
3.7 Auto-Cancel Booking	Cancels booking automatically if owner does not respond in 72 hours	72h timeout trigger	Cancel record → Booking DB

Data Stores used: Booking DB, Equipment DB, Audit Log

External Entities: Farmer, Equipment Owner

P3 Level 2 DFD



4.5.3 P4 — Escrow Management

This diagram shows how rental payments are tracked through the logical escrow system. It has 7 sub-processes:

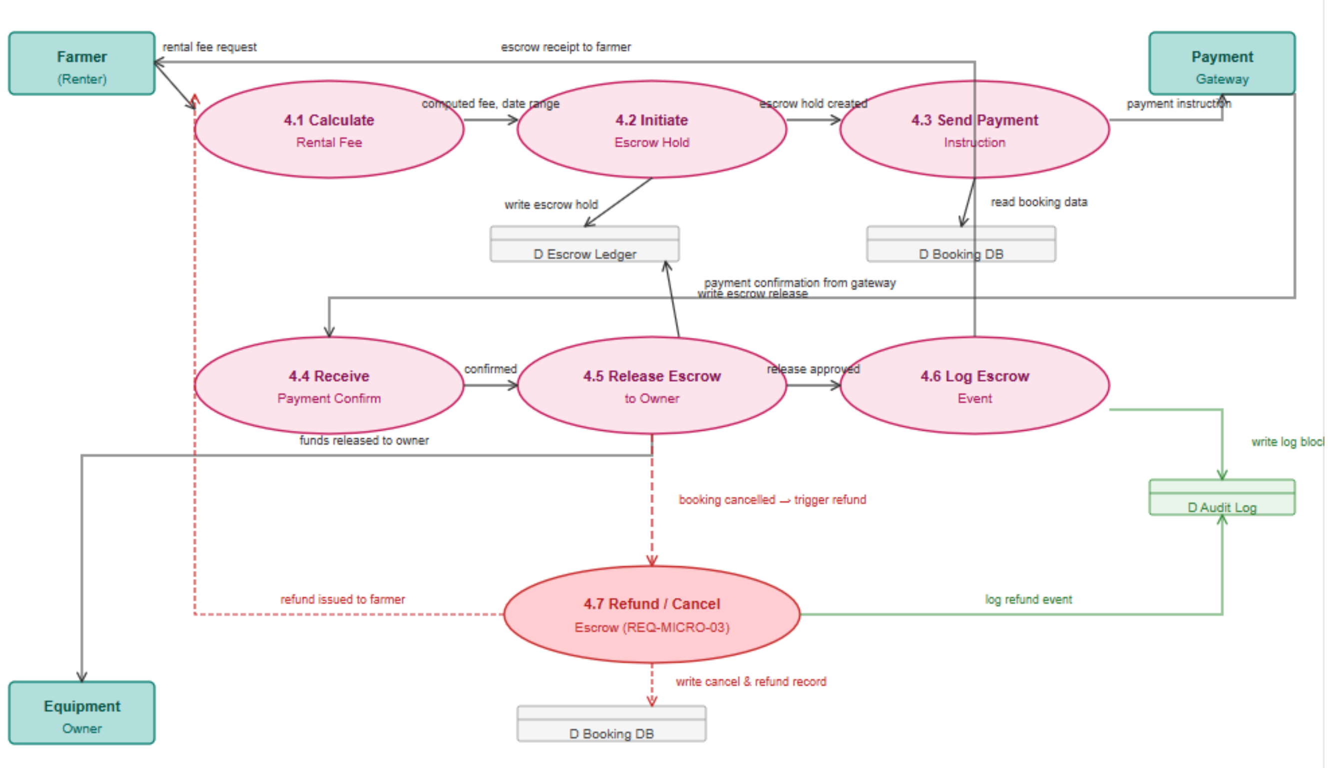
Sub-Process	What it Does	Input	Output
4.1 Calculate Rental Fee	Calculates total rental cost based on daily rate and number of days	Rental fee request	Computed fee + date range
4.2 Initiate Escrow Hold	Creates an escrow ledger entry in HELD state	Computed fee	Escrow hold written to Escrow Ledger
4.3 Send Payment Instruction	Sends payment details to the payment gateway	Escrow hold created	Payment instruction to gateway
4.4 Receive Payment Confirm	Receives payment confirmation from the gateway	Gateway confirmation	Confirmed payment status
4.5 Release Escrow to Owner	Updates escrow ledger to RELEASED when rental is complete	Confirmed + completed	Funds release record written
4.6 Log Escrow Event	Records all escrow actions in the audit log	Release approved	Log block → Audit Log
4.7 Refund / Cancel Escrow	Refunds farmer if booking is cancelled before rental starts	Booking cancelled trigger	Refund record + farmer notified

Data Stores used: Escrow Ledger, Booking DB, Audit Log

External Entities: Farmer, Equipment Owner, Payment Gateway

Note: No real money moves through the system — only payment status is tracked in the logical escrow ledger.

P4 Level 2 DFD


Farmer
(Renter)
rental fee request
escrow receipt to farmer
Payment
Gateway
4.1 Calculate
Rental Fee
computed fee, date range
4.2 Initiate
Escrow Hold
escrow hold created
4.3 Send Payment
Instruction
payment instruction
write escrow hold
read booking data
D Escrow Ledger
D Booking DB
payment confirmation from gateway
write escrow release
4.4 Receive
Payment Confirm
confirmed
4.5 Release Escrow
to Owner
release approved
4.6 Log Escrow
Event
funds released to owner
booking cancelled → trigger refund
D Audit Log
refund issued to farmer
4.7 Refund / Cancel
Escrow (REQ-MICRO-03)
log refund event
Equipment
Owner
write cancel & refund record
D Booking DB

4.5.4 P6 — Rental Inspection Workflow

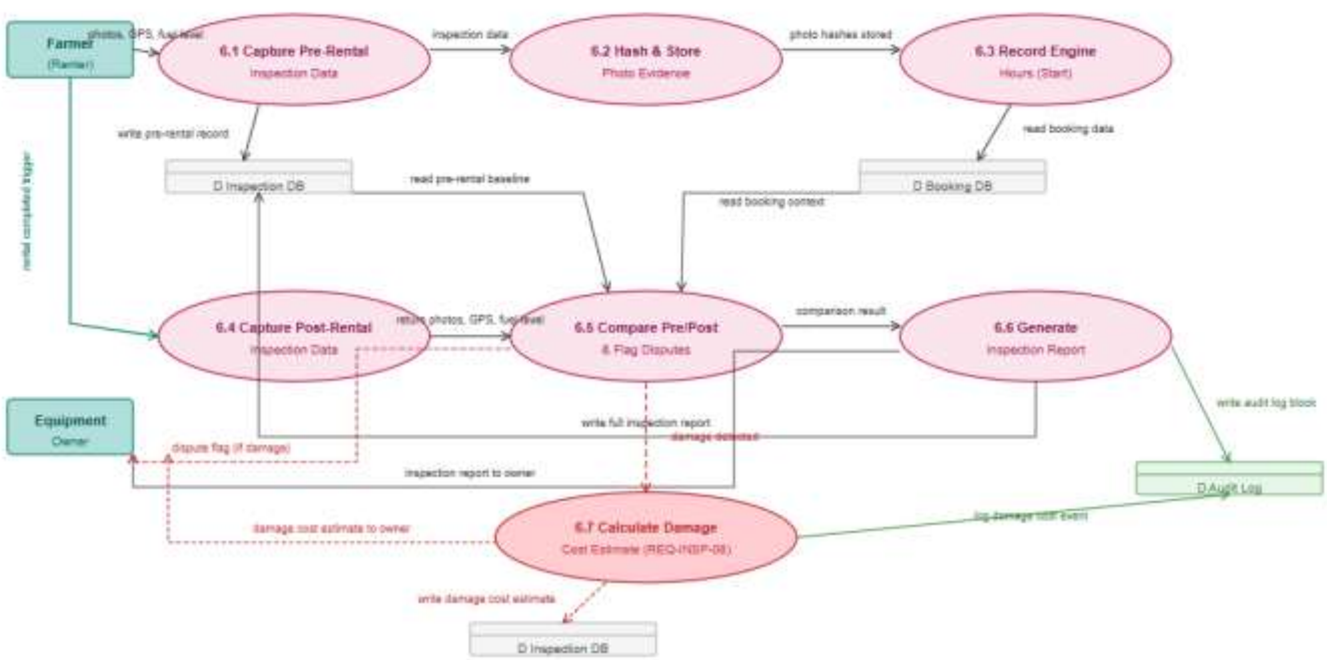
This diagram shows the full inspection process covering both pre-rental and post-rental inspections. It has 7 sub-processes:

Sub-Process	What it Does	Input	Output
6.1 Capture Pre-Rental Inspection Data	Records photos, GPS location, and fuel level before handover	Photos, GPS, fuel level	Pre-rental inspection data
6.2 Hash & Store Photo Evidence	Creates SHA-256 hashes of photos and stores them	Inspection data	Photo hashes stored → Inspection DB
6.3 Record Engine Hours (Start)	Records starting engine hours as a baseline for usage tracking	Engine hours reading	Baseline saved → Booking DB
6.4 Capture Post-Rental Inspection Data	Records the same data after equipment is returned	Return photos, GPS, fuel	Post-rental inspection data
6.5 Compare Pre/Post & Flag Disputes	Compares pre and post inspection to detect new damage	Pre + post data	Comparison result; dispute flag if damage
6.6 Generate Inspection Report	Creates a complete inspection report linked to the booking	Comparison result	Full inspection report → Inspection DB
6.7 Calculate Damage Cost Estimate	Estimates repair cost if damage is detected	Damage detected	Damage cost estimate → Inspection DB

Data Stores used: Inspection DB, Booking DB, Audit Log

External Entities: Farmer, Equipment Owner

P6 Level 2 DFD



5. Domain Model

The Domain Model shows the 16 main classes of the AgroFlow system, their key attributes, and how they relate to each other.

5.1 The 16 Domain Classes

ID	Class Name	Role in System
C-01	**User**	Base class — stores hashed CNIC, KYC status, and trust score.
C-02	**Farmer**	Extends User — land operators who rent equipment.
C-03	**Equipment Owner**	Extends User — owners who list machinery for rent.
C-04	**Administrator**	Extends User — approves KYC, resolves disputes, manages platform.
C-05	**Asset**	Represents a physical agricultural machine.
C-06	**Technical Specification**	Technical details of an asset (HP, fuel type, service date).
C-07	**Booking Engine**	Manages booking state transitions.
C-08	**Booking Record**	Records rental agreement between Farmer and Owner for an Asset.
C-09	**Booking Lifecycle**	Defines 10 valid states and transitions for a booking.
C-10	**Escrow Ledger**	Tracks payment hold and release per booking (logical only).
C-11	**Forensic Log Block**	Immutable audit record with hash-chain for tamper evidence.
C-12	**Inspection Report**	Photo-based condition report taken before and after each rental.
C-13	**Dispute Record**	Formal dispute linked to a booking — stores evidence and ruling.
C-14	**Geographic Search Engine**	Finds and ranks available assets by location.
C-15	**Asset Registry**	Data access layer for asset queries and registration.
C-16	**Telematics Event**	Real-time IoT data from a device attached to an asset. (Phase 2)

5.2 Key Relationships

Class A	Class B	Type	Meaning
User	Farmer / Owner / Admin	Generalisation	User is the parent class of all three roles
Equipment Owner	Asset	Composition	Owner owns one or more assets
Asset	Technical Specification	Composition	Each asset has exactly one set of technical specs
Booking Record	Escrow Ledger	Composition	Each booking has one escrow entry
Booking Record	Inspection Report	Aggregation	Each booking has pre and post inspection reports

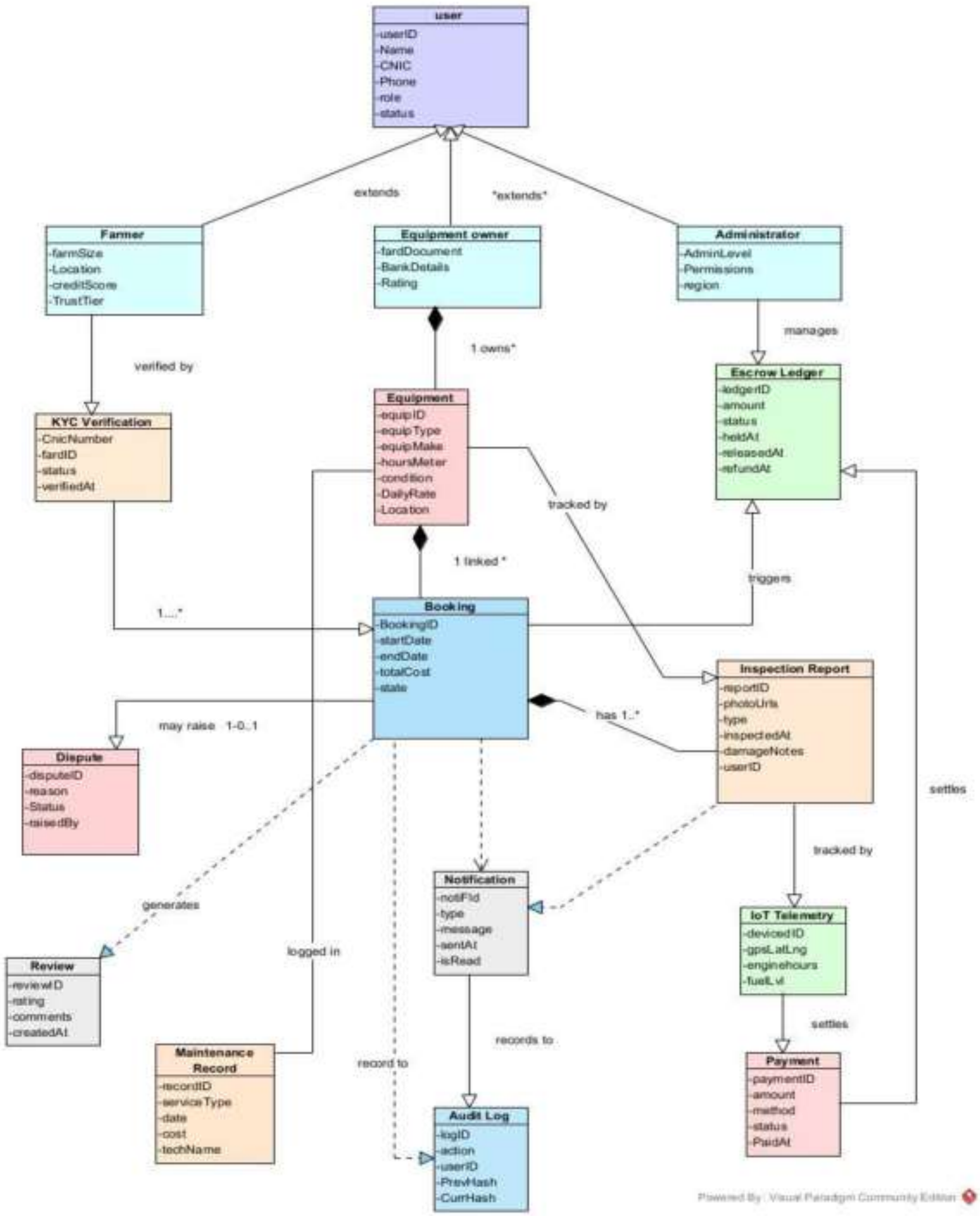
Class A	Class B	Type	Meaning
Booking Record	Dispute Record	Association	A booking may have one dispute
Booking Engine	Booking Lifecycle	Dependency	Engine uses lifecycle to manage state transitions
Forensic Log Block	Forensic Log Block	Self-Association	Each log block links to the previous one (hash chain)

5.3 Booking Lifecycle — 10 States

#	State	What it means
1	**REQUESTED**	Owner must confirm or reject within 24 hours.
2	**CONFIRMED**	Pre-rental inspection must be completed before handover.
3	**IN PROGRESS**	Equipment is in active use by the Farmer.
4	**PENDING RETURN**	Post-rental inspection must be submitted.
5	**COMPLETED**	No active dispute — escrow release is triggered.
6	**CANCELLED**	Cancelled before IN PROGRESS — refund triggered automatically.
7	**DISPUTED**	A dispute has been raised by either party.
8	**UNDER REVIEW**	Admin has been assigned and is reviewing evidence.
9	**RESOLVED**	Admin ruling issued — escrow action applied.
10	**CLOSED**	All actions complete — booking permanently closed.

5.4 Domain Model

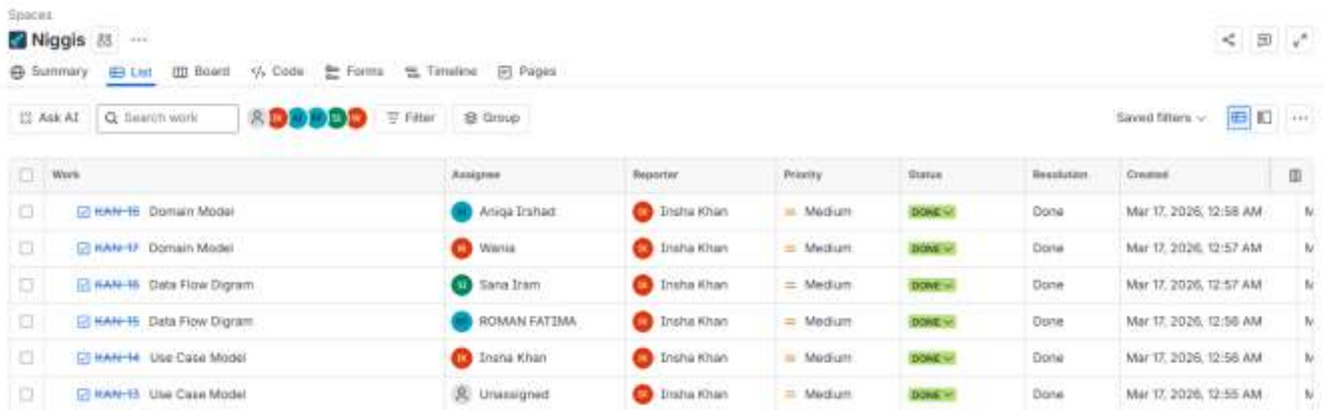
Domain Model:



6. Jira Board

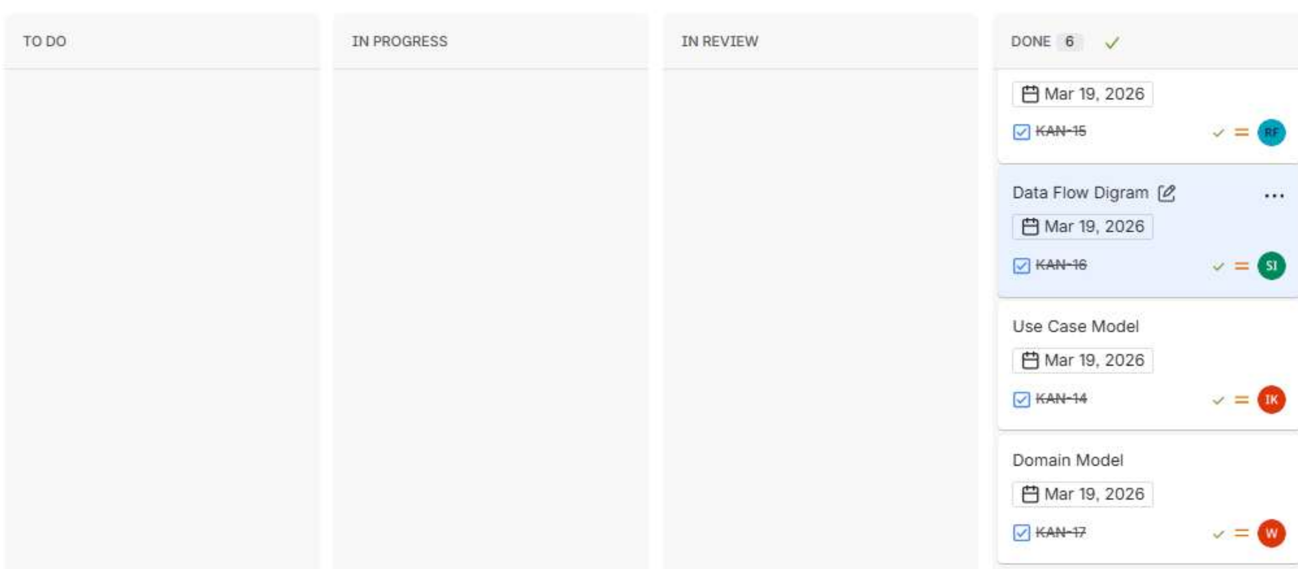
The screenshot below shows our Jira board with all tasks assigned to team members and their current status at the time of submission.

6.1 Sprint Task Summary

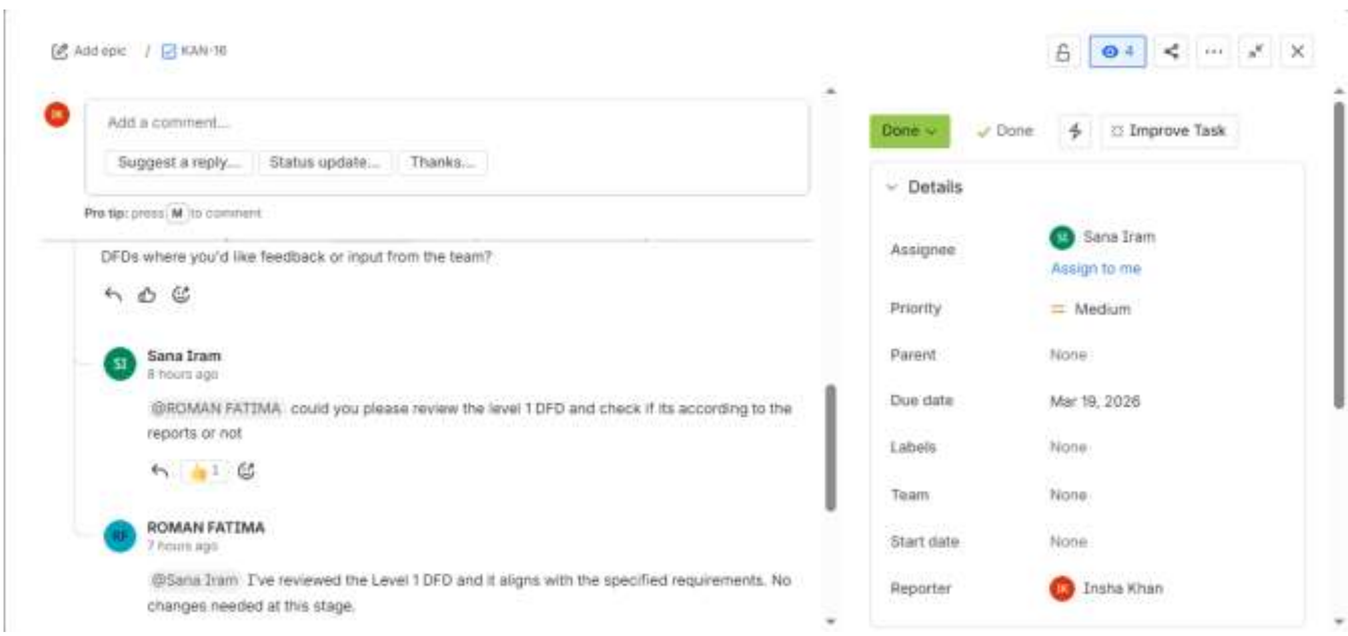


Work	Assignee	Reporter	Priority	Status	Resolution	Created
KAN-18 Domain Model	Aniqa Irshad	Insha Khan	Medium	DONE	Done	Mar 17, 2026, 12:58 AM
KAN-17 Domain Model	Wania	Insha Khan	Medium	DONE	Done	Mar 17, 2026, 12:57 AM
KAN-16 Data Flow Digram	Sana Iram	Insha Khan	Medium	DONE	Done	Mar 17, 2026, 12:57 AM
KAN-15 Data Flow Digram	ROMAN FATIMA	Insha Khan	Medium	DONE	Done	Mar 17, 2026, 12:56 AM
KAN-14 Use Case Model	Insha Khan	Insha Khan	Medium	DONE	Done	Mar 17, 2026, 12:56 AM
KAN-13 Use Case Model	Unassigned	Insha Khan	Medium	DONE	Done	Mar 17, 2026, 12:55 AM

Work	Assignee	Reporter	Priority	Status
KAN-18 Domain Model	Aniqa Irshad	Insha Khan	Medium	DONE
KAN-17 Domain Model	Wania	Insha Khan	Medium	DONE
KAN-16 Data Flow Digram	Sana Iram	Insha Khan	Medium	DONE
KAN-15 Data Flow Digram	ROMAN FATIMA	Insha Khan	Medium	DONE
KAN-14 Use Case Model	Insha Khan	Insha Khan	Medium	DONE
KAN-13 Use Case Model	Unassigned	Insha Khan	Medium	DONE



Sana’s



Roman’s

Add epic / KAN-15

Data Flow Digram

+

Description

level 0 Data Flow Diagram

level 2(1) DFD

Attachments 2

dfd level 2(1).jpeg
18 Mar 2026, 08:16 PM

DFD LEVEL0 .pdf
18 Mar 2026, 06:02 PM

Subtasks

Add subtask

Done | ✓ Done | Improve Task

Details

Assignee	ROMAN FATIMA Assign to me
Priority	Medium
Parent	None
Due date	Mar 19, 2026
Labels	None
Team	None
Start date	None
Reporter	Insha Khan

Add epic / KAN-15

Add a comment...

Suggest a reply... | Status update... | Thanks...

Pro tip: press M to comment

ROMAN FATIMA
7 hours ago

Level 2 — Process DFDs drill into the internals of each process:

- **P1 (UC1, UC13):** 6 sub-processes — collect data → validate CNIC → NADRA query → Fard verify → assign role/trust tier → log event

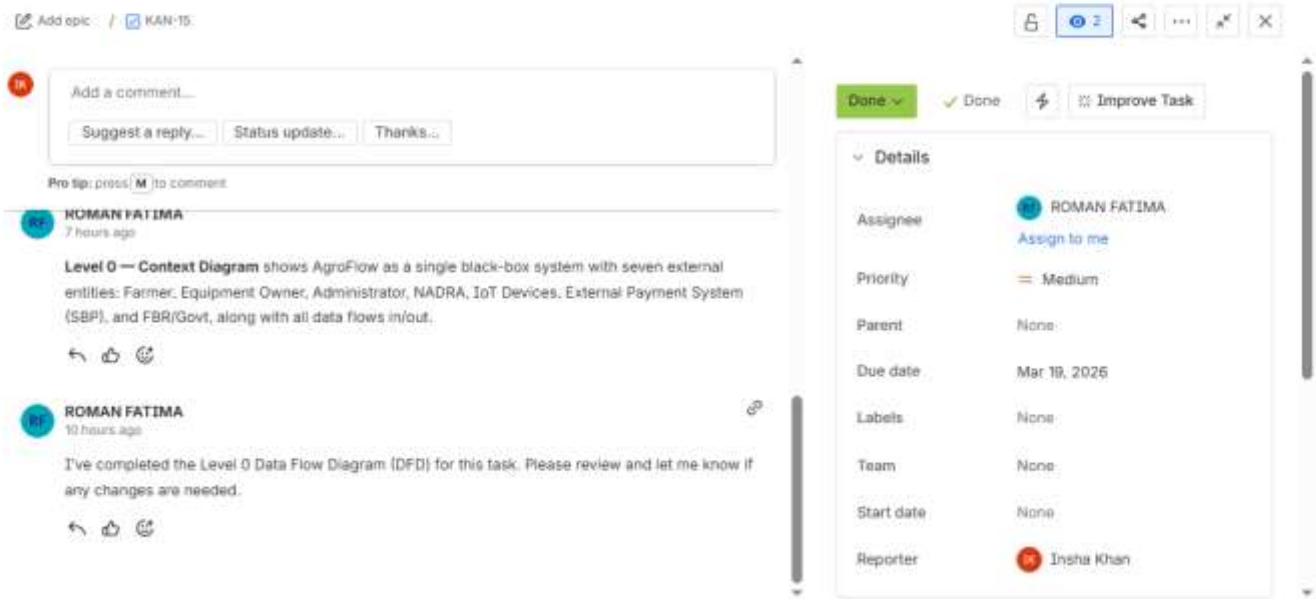
ROMAN FATIMA
7 hours ago

Level 0 — Context Diagram shows AgroFlow as a single black-box system with seven external entities: Farmer, Equipment Owner, Administrator, NADRA, IoT Devices, External Payment System (SBP), and FBR/Govt, along with all data flows in/out.

Done | ✓ Done | Improve Task

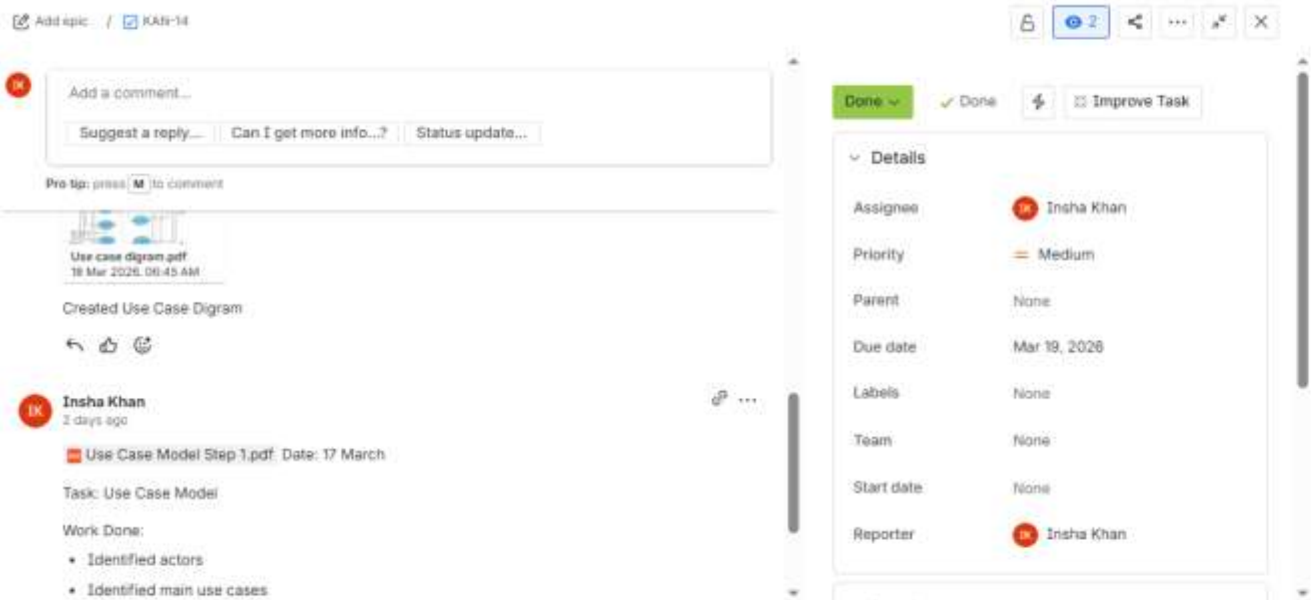
Details

Assignee	ROMAN FATIMA Assign to me
Priority	Medium
Parent	None
Due date	Mar 19, 2026
Labels	None
Team	None
Start date	None
Reporter	Insha Khan


Add epic / KAN-15
Add a comment...
Suggest a reply...
Status update...
Thanks...
Pro tip: press M to comment
ROMAN FATIMA
7 hours ago
Level 0 — Context Diagram shows AgroFlow as a single black-box system with seven external entities: Farmer, Equipment Owner, Administrator, NADRA, IoT Devices, External Payment System (SBP), and FBR/Govt, along with all data flows in/out.
ROMAN FATIMA
10 hours ago
I've completed the Level 0 Data Flow Diagram (DFD) for this task. Please review and let me know if any changes are needed.
Done
Done
Improve Task
Details
Assignee
ROMAN FATIMA
Assign to me
Priority
Medium
Parent
None
Due date
Mar 19, 2026
Labels
None
Team
None
Start date
None
Reporter
Insha Khan

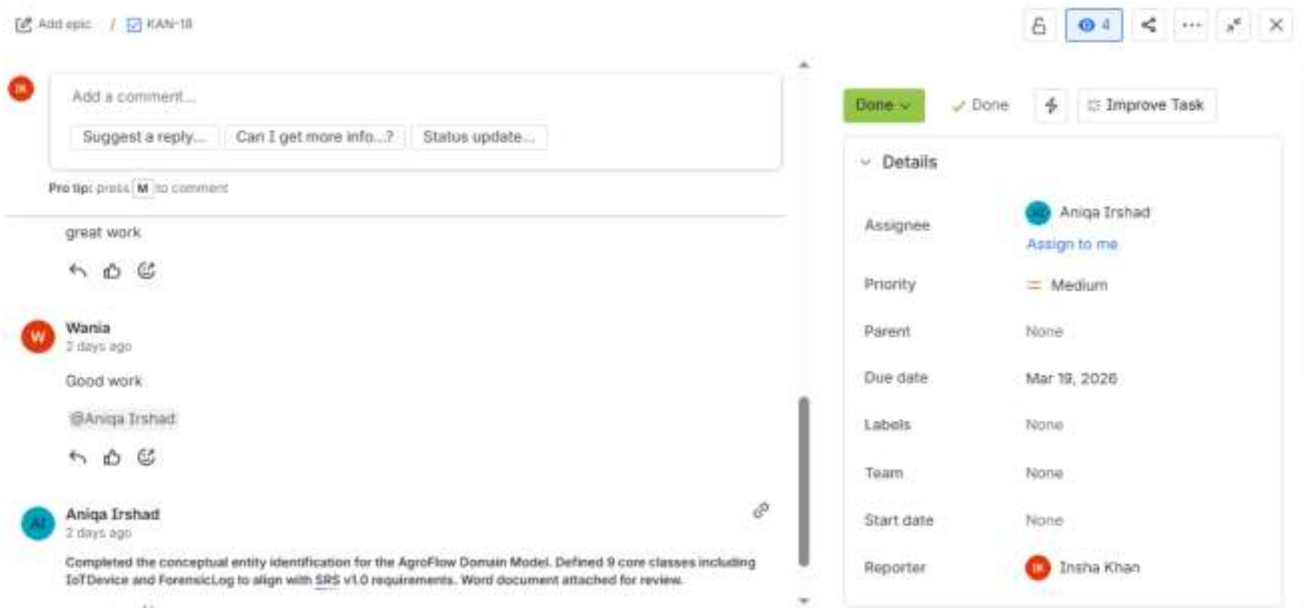
Insha's


Add epic / KAN-14
Use Case Model
Description
Add a description...
Attachments 3
Untitled-2nd us...ase.pdf
18 Mar 2026, 10:48 PM
Use case digram.pdf
18 Mar 2026, 06:46 AM
Use Case Mod... p 1.pdf
17 Mar 2026, 01:21 AM
Subtasks
Add subtask
Linked work items
Add linked work item
Done
Done
Improve Task
Details
Assignee
Insha Khan
Priority
Medium
Parent
None
Due date
Mar 19, 2026
Labels
None
Team
None
Start date
None
Reporter
Insha Khan


Add epic / KAN-14
Add a comment...
Suggest a reply...
Can I get more info...?
Status update...
Pro tip: press M to comment
Use case digram.pdf
Created Use Case Digram
Insha Khan
Use Case Model Step 1.pdf Date: 17 March
Task: Use Case Model
Work Done:
• Identified actors
• Identified main use cases
Done
Done
Improve Task
Details
Assignee
Insha Khan
Priority
Medium
Parent
None
Due date
Mar 19, 2026
Labels
None
Team
None
Start date
None
Reporter
Insha Khan

Aniqa’s


Add epic / KAN-18
Domain Model
Description
Add a description...
Attachments 2
Domain Model.jpg
Domain model.pdf
Subtasks
Add subtask
Linked work items
Add linked work item
Done
Done
Improve Task
Details
Assignee
Aniqa Irshad
Assign to me
Priority
Medium
Parent
None
Due date
Mar 19, 2026
Labels
None
Team
None
Start date
None
Reporter
Insha Khan


Add epic / KAN-18
Add a comment...
Suggest a reply... Can I get more info...? Status update...
Pro tip: press M to comment
great work
Wania
2 days ago
Good work
@Aniqa Irshad
Aniqa Irshad
2 days ago
Completed the conceptual entity identification for the AgroFlow Domain Model. Defined 9 core classes including IoTDevice and ForensicLog to align with SRS v1.0 requirements. Word document attached for review.
Done
Done
Improve Task
Details
Assignee
Aniqa Irshad
Assign to me
Priority
Medium
Parent
None
Due date
Mar 19, 2026
Labels
None
Team
None
Start date
None
Reporter
Insha Khan

Wania's


Add epic / KAN-17
Domain Model
Description
Made a draft
Attachments 1
Subtasks
Add subtask
Linked work items
Done
Done
Improve Task
Details
Assignee
Wania
Assign to me
Priority
Medium
Parent
None
Due date
Mar 19, 2026
Labels
None
Team
None
Start date
None
Reporter
Insha Khan


Add epic
KAN-17
Linked work items
Add linked work item
Activity
All
Comments
History
Work log
Add a comment...
Suggest a reply...
Status update...
Thanks...
Pro tip: press M to comment
Wania
2 days ago
on it
Done
Done
Improve Task
Details
Assignee
Wania
Assign to me
Priority
Medium
Parent
None
Due date
Mar 19, 2026
Labels
None
Team
None
Start date
None
Reporter
Insha Khan

THE END ☺